

D-caRbon: Geographically-Aware Building-to-Grid Carbon Mitigation Potential from Demand Response

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Abstract—Large energy-importing metropolitan areas face a distinctive decarbonisation challenge because a large share of their operational emissions is determined by upstream power generation outside the city boundary. In such cities, building cooling loads can provide indirect carbon mitigation through demand response, but their potential is strongly affected by spatially heterogeneous urban conditions. This paper proposes a geographically-aware framework that models thermostatically controlled cooling loads as equivalent energy storage and extends the assessment to urban building clusters by incorporating weather observations, building morphology, urban heat island effects, and air-conditioning heat-rejection feedback. The resulting flexibility is aggregated and embedded into a unit commitment model to estimate carbon-emission mitigation. A case study for Macau shows that flexibility and mitigation potential vary significantly across buildings and districts: large commercial and public buildings contribute disproportionately, while dense urban areas experience higher local temperature rise and altered electrical response through temperature-dependent air-conditioning efficiency. At the city scale, aggregation smooths individual variability and yields a predictable demand response envelope for power-system scheduling.

Index Terms—Smart building, demand response, decarbonisation, virtual energy storage

I. INTRODUCTION

Urban areas play a dominant role in global energy consumption and carbon emissions. Although cities occupy less than 2% of the global land area, they consume nearly 80% of energy and contribute more than 60% of global carbon emissions [1]. Reducing the carbon footprint of urban energy use is therefore central to climate mitigation. In many large metropolitan regions, especially energy-importing cities, electricity consumption is the dominant source of operational emissions. Because much of that electricity is imported, the associated emissions are largely determined by upstream generation rather than local activities, limiting the direct carbon benefits of conventional local efficiency measures.

Despite this constraint, energy-importing cities also possess substantial indirect decarbonisation potential through flexible demand-side resources. The building sector typically accounts for a major share of urban electricity demand, and space cooling often makes up a large fraction of summer peak load. Building thermal inertia and the controllability of thermostatically controlled loads (TCLs), such as air-conditioning systems

and heat pumps, allow electricity demand to be shifted or reduced for short periods without materially compromising occupant comfort. Through demand response (DR), this flexibility can support power-system operation, improve renewable-energy integration, and indirectly reduce system-wide carbon emissions.

Existing studies approach this issue from two related but only partially connected perspectives. In the energy and power literature, controllable air-conditioning and other TCLs have been analysed as flexible resources for balancing, scheduling, and frequency regulation, and have been modelled as equivalent storage or broader building-to-grid resources [2]–[6]. These ideas have also been demonstrated in practical programs and pilot projects, including ancillary-service participation and coordinated control of building clusters [7]–[9]. In the building environment literature, attention has focused more on operational carbon accounting, low-carbon building performance, and coordinated control of buildings and distributed energy systems [10]–[14]. This work offers valuable insight into building-level performance, but usually places less emphasis on the system-level carbon value created when buildings interact with the wider grid.

Taken together, these two strands reveal a disciplinary gap. The energy and power literature emphasizes electrical coordination but often simplifies the geographical and microclimatic heterogeneity that shapes urban cooling demand, whereas the building environment literature better captures building characteristics and direct decarbonisation measures but pays less attention to system-level carbon mitigation from flexibility. Consequently, the carbon-mitigation potential of coordinated building-grid interaction remains insufficiently quantified at both city and building scales.

Addressing this gap requires integrating power-system operation, building thermodynamics, urban climate modelling, and geospatial analysis. Spatial heterogeneity in building density, urban heat island intensity, and local microclimate affects cooling demand, HVAC efficiency, and thus urban building flexibility; neglecting these factors can bias estimates of demand-response capability and carbon-mitigation potential.

This paper therefore proposes a geographically-aware framework for assessing the carbon-mitigation potential of building-level demand response. The framework combines

building thermal flexibility modelling, urban microclimate estimation, and spatial aggregation to evaluate the equivalent energy-storage capacity and demand-response envelope of large-scale urban building clusters. Its contributions are: 1) a building-level demand-response model that represents TCLs as equivalent energy storage while preserving key thermal dynamics and comfort constraints; 2) a geographically-aware urban modelling approach integrating building morphology, weather observations, urban heat island effects, and air-conditioning heat-rejection feedback; and 3) a quantitative basis for evaluating the role of urban buildings in low-carbon power-system operation and demand-response planning.

II. DEMAND RESPONSE ENVELOPE OF A SINGLE BUILDING

When a building provides demand response services, its thermal behaviour can be represented as an equivalent energy storage system [15]. The building is modelled as a single thermal zone with uniform indoor temperature. Under a first-order lumped-capacitance assumption, the indoor thermal dynamics are

$$C_a \rho_a V_b \frac{dT_b^r}{dt} = (T^a - T_b^r) K_b A_b^s + C_a \rho_a V_b (T^a - T_b^r) N_b^{ex} + \varepsilon_b A_b + I_b^{\text{solar}} A_b^{\text{sol}} - Q_b^{\text{cool}}, \quad (1)$$

where the terms on the right-hand side denote, respectively, conductive heat transfer through the envelope, heat exchange due to ventilation and infiltration, internal heat gains, solar heat gains, and cooling supplied by the TCL. Here, T_b^r is the indoor temperature of building b , T^a is the ambient temperature, K_b is the overall heat-transfer coefficient, A_b^s is the exposed surface area, N_b^{ex} is the air-exchange rate, $\varepsilon_b A_b$ is the internal heat gain, I_b^{solar} is the incident solar radiation density, and A_b^{sol} is the effective solar receiving area. The electrical power consumption of the TCL is obtained from the cooling rate through the coefficient of performance (COP).

During normal operation, the indoor temperature is maintained at the comfort setpoint $T_b^{r,\text{comf}}$. For cooling-dominated conditions, the building is assumed to be in thermal steady state under baseline control, namely $dT_b^r/dt = 0$. The baseline electrical power consumption of the TCL is then

$$P_b|_{T_b^r=T_b^{r,\text{comf}}} = \text{COP}_b^{-1} \left((T^a - T_b^{r,\text{comf}}) K_b A_b^s + C_a \rho_a V_b (T^a - T_b^{r,\text{comf}}) N_b^{ex} + \varepsilon_b A_b + I_b^{\text{solar}} A_b^{\text{sol}} \right) \quad (2)$$

where COP_b is the coefficient of performance of the TCL. At the comfort setpoint, the baseline cooling demand exactly balances the total indoor heat gain.

When providing demand response, the TCL cooling power is adjusted around its baseline operating point. If cooling is reduced temporarily, the indoor temperature rises toward the upper comfort bound and the corresponding electrical power is denoted by $P_b|_{T_b^r=T_b^{r,\text{max}}}$. Conversely, if the TCL is temporarily over-supplied, the indoor temperature decreases toward the

lower comfort bound and the corresponding electrical power is denoted by $P_b|_{T_b^r=T_b^{r,\text{min}}}$.

Under this interpretation, the building can be regarded as an equivalent energy storage device. Additional cooling that moves the indoor temperature from $T_b^{r,\text{comf}}$ to $T_b^{r,\text{min}}$ is interpreted as charging, while reduced cooling that allows the temperature to increase from $T_b^{r,\text{comf}}$ to $T_b^{r,\text{max}}$ is interpreted as discharging. The equivalent charging and discharging power, denoted by P_b^{char} and P_b^{disc} , are therefore given by the deviation of TCL power from the baseline:

$$P_b^{\text{char}} = P_b|_{T_b^r=T_b^{r,\text{min}}} - P_b|_{T_b^r=T_b^{r,\text{comf}}} = \text{COP}_b^{-1} \left((T_b^{r,\text{comf}} - T_b^{r,\text{min}}) K_b A_b^s + C_a \rho_a V_b (T_b^{r,\text{comf}} - T_b^{r,\text{min}}) N_b^{ex} \right) \quad (3)$$

$$P_b^{\text{disc}} = P_b|_{T_b^r=T_b^{r,\text{max}}} - P_b|_{T_b^r=T_b^{r,\text{comf}}} = \text{COP}_b^{-1} \left((T_b^{r,\text{max}} - T_b^{r,\text{comf}}) K_b A_b^s + C_a \rho_a V_b (T_b^{r,\text{max}} - T_b^{r,\text{comf}}) N_b^{ex} \right) \quad (4)$$

where $T_b^{r,\text{min}}$ and $T_b^{r,\text{max}}$ are the lower and upper comfort bounds, respectively. Under this sign convention, both P_b^{char} and P_b^{disc} are non-negative and represent the upward and downward flexibility margins of the TCL relative to baseline operation.

The capacity of the equivalent energy storage is

$$E_b^{\text{eq}} = \text{COP}_b^{-1} C_a \rho_a V_b (T_b^{r,\text{max}} - T_b^{r,\text{min}}) \quad (5)$$

The absolute electrical power consumption of the TCL depends on ambient temperature through the thermal balance equation. By contrast, the equivalent charging power, discharging power, and storage capacity above depend only on the comfort band and intrinsic thermal characteristics of the building. In particular, the storage capacity in (5) is determined by the allowable indoor temperature range and the effective thermal mass represented by the room air volume.

For short-term flexibility analysis of a single building, COP_b can be treated as constant over the demand-response interval. At the urban scale, however, the COP should be updated according to building use type and ambient temperature. For each building i , let $\text{COP}^{\text{base}}(u_i)$ denote the base COP associated with use type u_i . A linearly corrected COP is first computed and then truncated to a physically admissible interval:

$$\text{COP}_i = \min\{\text{COP}_{\text{max}}, \max\{\text{COP}_{\text{min}}, \text{COP}^{\text{base}}(u_i) - k_{\text{cop}}(\bar{T}_i - T_{\text{ref}})\}\}, \quad (6)$$

where $k_{\text{cop}} = 0.05 \text{ K}^{-1}$, $T_{\text{ref}} = 30^\circ \text{C}$, $\text{COP}_{\text{min}} = 2.5$, and $\text{COP}_{\text{max}} = 4.0$. This expression keeps the temperature-corrected COP within the prescribed bounds and is used in the subsequent urban-scale assessment to update the electrical flexibility of individual buildings under spatially varying microclimatic conditions.

III. DEMAND RESPONSE ENVELOPE OF URBAN BUILDING CLUSTERS CONSIDERING MICROCLIMATE

At the urban scale, building flexibility cannot be obtained by simply summing single-building results, because spatially varying microclimatic conditions alter the local ambient temperature and hence the electrical flexibility of cooling systems through temperature-dependent COP. We therefore discretise the urban area into square grids of side length L_g to correct the local temperature seen by each building.

The baseline ambient temperature is derived from weather-station observations by spatial weighting. For building i , let $T_s(t)$ denote the temperature recorded at station s and let d_{is} be the Euclidean distance to that station. The station weights, temperature series, and mean ambient temperature are

$$\begin{aligned} w_{is} &= \frac{d_{is}^{-2}}{\sum_{s'=1}^{N_s} d_{is'}^{-2}}, \\ T_i(t) &= \sum_{s=1}^{N_s} w_{is} T_s(t), \\ \bar{T}_i &= \frac{1}{N_t} \sum_{t=1}^{N_t} T_i(t), \end{aligned} \quad (7)$$

where N_s is the number of weather stations and N_t is the number of time samples. The mean value is taken as the baseline ambient temperature, denoted by T_z^{met} for the corresponding grid.

The local temperature correction consists of an urban heat island component and an air-conditioning heat-rejection component. The building density of grid z is

$$D_z = \frac{n_z}{A_z} \times 10^6 \quad [\text{buildings/km}^2], \quad (8)$$

where n_z is the number of buildings in the grid and A_z is the grid area in m^2 . The UHI increment is represented by piecewise-linear interpolation,

$$\begin{aligned} \Delta T_z^{\text{UHI}} &= \mathcal{I}(D_z; [0, q_{1/3}, q_{1/2}, q_{2/3}], \\ &\quad [\Delta T_{\text{open}}, \Delta T_{\text{open}}, \Delta T_{\text{normal}}, \Delta T_{\text{dense}}]), \end{aligned} \quad (9)$$

where $q_{1/3}$, $q_{1/2}$, and $q_{2/3}$ are the positive-density quantiles defining open, normal-urban, and dense-urban anchors.

Using the equivalent discharging power as the cooling-side response, the rejected heat of building i is

$$Q_i^{\text{rej}} = P_i^{\text{dis}} \left(1 + \frac{1}{\text{COP}_i} \right). \quad (10)$$

At the grid level, the associated AC-induced temperature rise is

$$\Delta T_z^{\text{AC}} = k_z \frac{\sum_{i \in z} Q_i^{\text{rej}}}{A_z}, \quad (11)$$

where k_z is a heat-dispersion coefficient selected from $\{k_{\text{open}}, k_{\text{normal}}, k_{\text{dense}}\}$ or, when physical parameters are available, computed as $k_z = (\rho_a c_a h_z^{\text{mix}} u_z^{\text{eff}})^{-1}$. The resulting local grid temperature is

$$T_z^{\text{local}} = T_z^{\text{met}} + \Delta T_z^{\text{UHI}} + \Delta T_z^{\text{AC}}. \quad (12)$$

This updated temperature is fed back to recompute only COP_i , comfort variables, E_i^{eq} , P_i^{ch} , and P_i^{dis} , while geometry and envelope parameters remain unchanged.

The aggregated flexibility metrics are

$$E^{\text{agg}} = \sum_{i=1}^{N_b} E_i^{\text{eq}}, \quad P^{\text{agg}} = \sum_{i=1}^{N_b} P_i^{\text{dis}}. \quad (13)$$

The aggregated equivalent model is then incorporated into a unit commitment model as a reserve resource. Let the optimal generation dispatch without and with building demand response be denoted by $\{P_g^{\text{gen}}\}$ and $\{P_g^{\text{gen,DR}}\}$, respectively. The marginal carbon-emission mitigation is calculated as

$$\Delta C^{\text{mit}} = \sum_{g=1}^{N_g} (P_g^{\text{gen}} - P_g^{\text{gen,DR}}) \gamma_g. \quad (14)$$

IV. CASE STUDIES

This section evaluates the proposed geographically-aware framework in an urban Macau setting and quantifies how microclimate corrections and building-level heterogeneity affect demand-response-based carbon mitigation outcomes.

Weather data are obtained from the Macau Meteorological and Geophysical Bureau [16]. The dataset contains hourly temperature, humidity, solar radiation, and wind speed from 9 weather stations during the summer of 2023. Building geometry data are obtained from OpenStreetMap [17]; missing heights are imputed using KNN and supplemented by a commercial dataset. Thermal envelope parameters are assigned according to building use type and construction vintage, while UHI and heat-dispersion parameters are calibrated from local temperature and building-density data. Power-system carbon-emission factors are taken from the PRC Ministry of Ecology and Environment [18], and the unit commitment model is implemented on the IEEE 24-bus system in PyPower.

A. Individual building flexibility

Four representative buildings are selected from the commercial, public, residential, and industrial sectors, as shown in Fig. 1. All four exhibit substantial flexibility, but the public building (Macau East Asian Games Dome) shows the largest response, with a maximum discharging power of about 110 kW and an equivalent storage capacity of about 200 kWh, mainly because of its large floor area. The residential building (Nova City) provides smaller peak flexibility, around 40 kW, but can sustain the response for longer because of its better thermal insulation. Time variation in electrical flexibility is mainly driven by COP variation under changing microclimatic conditions.

B. Clustered building flexibility

At the cluster scale, spatial heterogeneity matters. The strongest microclimate effect appears in the central Macau Peninsula, where dense urban form and intensive air-conditioning use produce a local temperature rise of about 1.6°C , corresponding to an estimated 4.2% COP reduction. By contrast, most of Taipa and Coloane experience much smaller

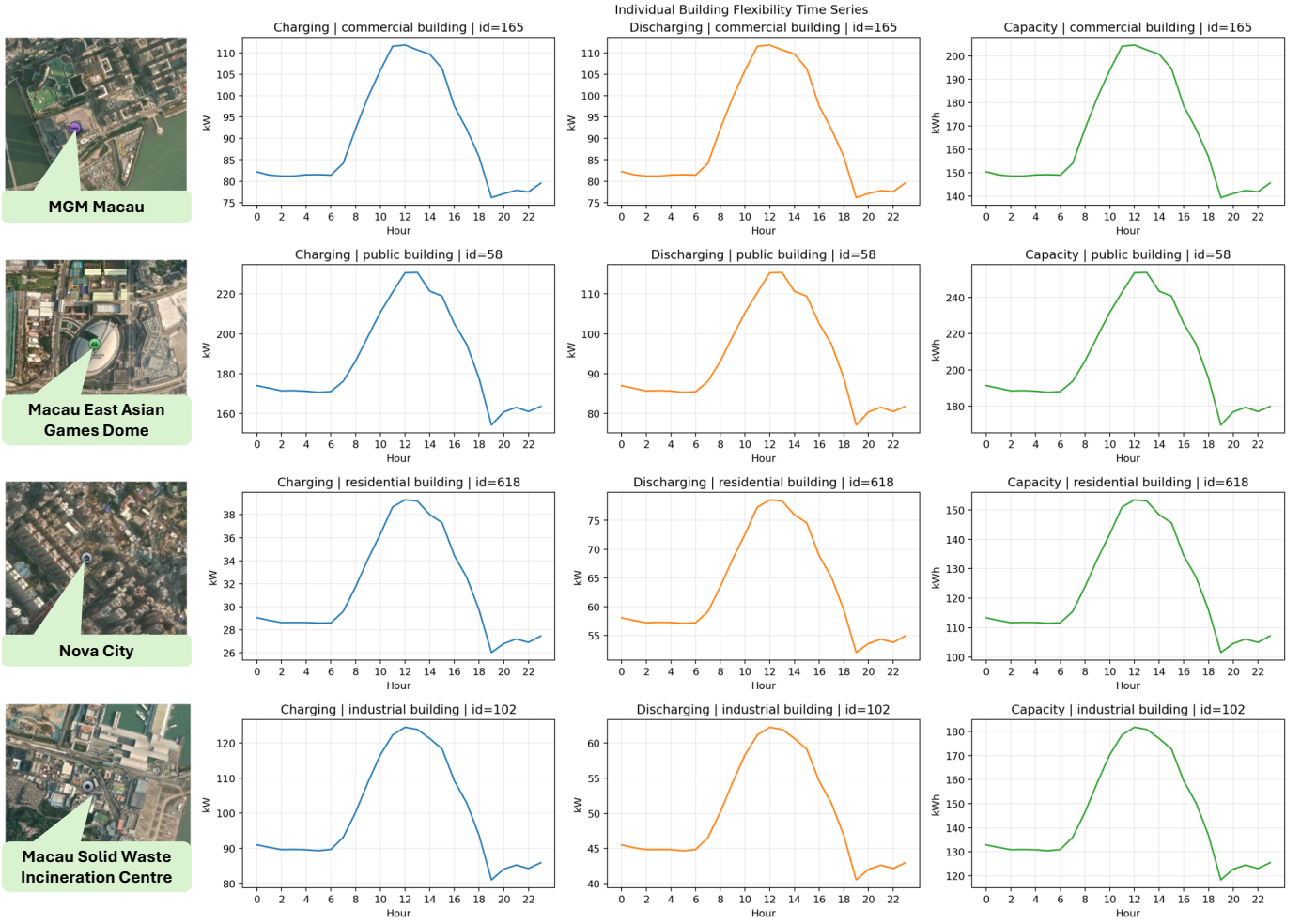


Fig. 1. Example of building-level demand response flexibility.

temperature increases, around 0.4°C and 0.2°C , respectively, because of lower density and better air exchange with the sea, as shown in Fig. 2.

At the system level, Fig. 3 shows that Macau can provide a maximum equivalent discharging power of about 68 MW. This value is lower than in studies based on full air-conditioning interruption, but it is compatible with sustained operation under comfort constraints. The microclimate effect increases the total demand-response potential by approximately 73 MWh, with the largest increment occurring during daytime hours when cooling demand is highest.

C. Carbon emission mitigation map

By flexibly participating in wider power-system operation, demand response from urban building clusters can reduce carbon emissions by displacing fossil-fuel-based generation during peak demand periods. The spatial distribution of carbon-mitigation potential is visualised in Fig. 4, and an interactive version can be accessed in the GitHub repository [19]. The deepest red areas indicate higher carbon-mitigation potential, but these areas do not always coincide with the highest-density

districts identified in the previous section. In the centre of the Macau Peninsula, for example, building density is high, but the carbon-mitigation potential is not as large as might be expected because many of the buildings are relatively small residential units. Their collective contribution should nevertheless be recognised. Some residential buildings still stand out because of their exceptional height, but buildings with large volume and high thermal capacity, such as MGM Macau, remain the most prominent contributors. By aggregating the carbon-mitigation potential of all buildings in Macau, the total carbon-emission reduction is estimated to be around 83.9 tons of CO_2 per year, equivalent to 24% of Macau's total carbon emissions from cooling. This result highlights the important role that building-level demand response can play in supporting low-carbon energy systems and advancing urban sustainability goals.

V. CONCLUSIONS

This paper proposes a geographically-aware framework for assessing the carbon-mitigation potential of building-to-grid demand response in dense urban environments. By combining

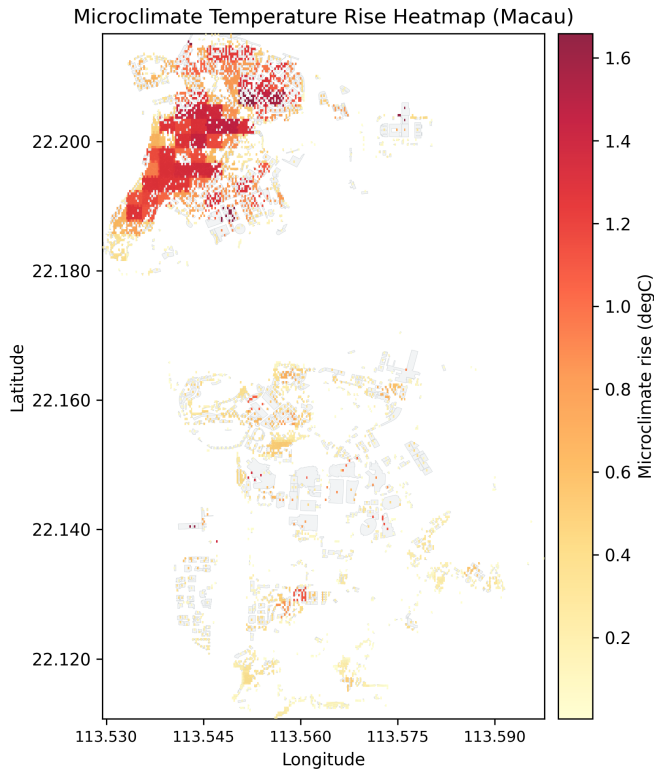


Fig. 2. Estimated local temperature increase due to urban heat island effects and air-conditioning-induced heat rejection across the urban area of Macau.

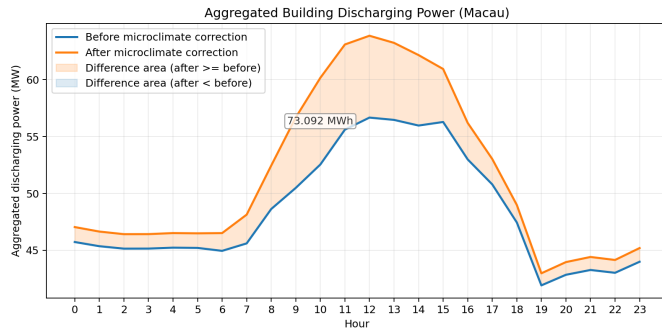


Fig. 3. Time series of aggregated equivalent discharging power from the urban building cluster during the summer period.

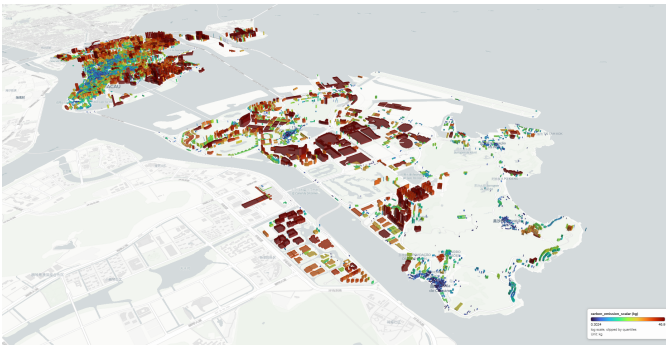


Fig. 4. Spatial distribution of carbon emission mitigation potential across the urban area.

an equivalent energy-storage representation of building cooling loads with spatially resolved microclimate corrections, the framework captures both building-level heterogeneity and city-scale aggregation effects. The Macau case study indicates that large commercial and public buildings provide substantial flexibility, while microclimate-induced COP variation materially affects the electrical demand-response potential and the resulting carbon-mitigation outcome. These findings support the use of geographically informed demand-response assessment for urban energy planning and low-carbon power-system operation.

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